

Regulating Uncertainty; Systemic Risk in the AI Act

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WASP—HS

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A premise

“...we have not given nuclear power generation systems enough time to express themselves...we are only just beginning to uncover the potential dangers that make any prediction of risk very uncertain.”

Charles Perrow, *Normal Accidents*, 2011

Outline

- Motivation
- Background
- Systemic Risk
- Legal Mechanisms
- Three core problems

Motivation

- Critically analyze the concept of systemic risk within the AI Act;
 - Does systemic risk generate more uncertainty than it resolves?
- Highlight some of the problems with flexible regulation as a means of avoiding the pacing problem.

Background

- Horizontal regulation, coming into force iteratively, began legally binding on 1 August 2024
- Dual purpose: safeguard European values and fundamental rights, promote innovation.



Systemic Risk

- A new concept (OECD 2003 report)
- Risk of breakdown in system, rather than just its parts, and is seen in the "co-movements" between system parts [Kaufmann and Scott 2003]
- Four components
 - Complexity
 - Uncertainty
 - Ambiguity
 - Ripple effects beyond the source of the risk

Legal mechanisms

- High-impact capabilities
- Computational thresholds
 - Exceeding 10^{25} FLOP
- Ex officio decisions, implementing acts, delegated acts

Three Core Critiques

1. The Institutionalisation of uncertainty

- Law relies on science to define risk, but in the case of GPAI, there is a lack of consensus regarding the risks
- Regulatory framework solidifies an inherently unstable concept

2. The delegation of systemic risk to Model providers

- Risk of regulatory capture in the operationalisation of systemic risk

3. The creation of uncertainty between model and system.

- The Act draws a line between GPAI and AI systems (for legal certainty)
- Systems instantiate the risks embedded in models – yet systemic risk obligations are not extended to systems

Summary

- Issues with the use of technical standards such as compute as a way of determining systemic risk.
- Tension between innovation and fundamental rights

Future research?

- Evolution of systemic risk within the AI Act's regulatory framework
 - Analysing the shifts in systemic risk through the different drafts of the Code of Practice

Thank you!

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